Hunmin (Troy) Lee

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Education

Ph.D. **University of Minnesota**, Computer Science and Engineering — *Twin Cities, MN, USA*, Aug. 2022 to Current
- Advisor: Dr. Catherine Qi Zhao
- GPA: 3.97/4.0

MS **Georgia State University**, Computer Science — *Atlanta, GA, USA*, Aug. 2020 to May 2022
- Advisor: Dr. Donghyun Kim and Dr. Yingshu Li
- GPA: 4.11/4.3

BE **Chonnam National University**, Computer Engineering — *Gwangju, South Korea*, May 2014 to Feb. 2020
- Advisor: Dr. Youngchul Kim
- GPA: 3.83/4.5

Publications (Journals)

[J.16] **Hunmin Lee**, Brian Lim, Ming Jiang, Zhi Yang, Qi Zhao. Few-Shot Prototype Adaptation for Generalizable Electromyography Gesture Recognition. *Scientific Reports* (**IF: 3.9**, at publication) — Mar. 2026

[J.15] **Hunmin Lee**, Kehua Wang, Daehee Seo. FedPML: Federated Prototype Meta-Learning for Adaptation and Generalization in Distributed Environments. *IEEE Internet of Things Journal*, vol. 13, no. 6, pp. 11361-11372. (**IF: 9.936**, at publication) — Jan. 2026

[J.14] **Hunmin Lee**, Daehee Seo. Federated Prototype Adaptation Learning: Personalized and Generalizable Learning in Distributed IoT Networks. *IEEE Internet of Things Journal*, vol. 13, no. 2, pp. 2408-2425. (**IF: 9.936**, at publication) — Nov. 2025

[J.13] **Hunmin Lee**, Ming Jiang, Zhi Yang, Qi Zhao. FedEMG: Achieving Generalization, Personalization, and Resource Efficiency in EMG-based Upper-Limb Rehabilitation through Federated Prototype Learning. *IEEE Transactions on Biomedical Engineering*, vol. 73, no. 2, pp. 803-817. (**IF: 4.5**, at publication) — Jul. 2025

[J.12] **Hunmin Lee**, Hongju Seong, Jihun Bae, Wonbin Kim, Hyeokchan Kwon, Daehee Seo. FedVaccine: Robust Federated Learning in Noisy and Non-IID Wireless Network Environments. *IEEE Transactions on Network Science and Engineering*, vol. 13, pp. 701-716. (**IF: 7.9**, at publication) — Jul. 2025

[J.11] **Hunmin Lee**, Ming Jiang, Jinhui Yang, Zhi Yang, Qi Zhao. Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 404-419. (**IF: 5.2**, at publication) — Dec. 2024

[J.10] **Hunmin Lee**, Daehee Seo. CLSM-FL: Clustering-based Semantic Federated Learning in non-IID IoT Environment. *IEEE Internet of Things Journal*, vol. 11, no. 18, pp. 29838-29851. (**IF: 9.936**, at publication) — Jul. 2024

[J.9] **Hunmin Lee**, Ming Jiang, Jinhui Yang, Zhi Yang, Qi Zhao. Unveiling EMG semantics: a prototype-learning approach to generalizable gesture classification. *Journal of Neural Engineering*, vol. 21, no. 3, 036031. (**IF: 3.8**, at publication) — May. 2024

[J.8] **Hunmin Lee**, Jihun Bae, Hyeongok Lee. Towards Utilization of Explainable Hand Motion Recognition in 3D Capacitive Proximity Sensing. *IEEE Sensors Journal*, vol. 24, no. 12, pp. 20076-20091. (**IF: 4.5**, at publication) — Apr. 2024

[J.7] **Hunmin Lee**, Daehee Seo. FedLC: Optimizing federated learning in non-IID data via label-wise clustering. *IEEE Access*, vol. 11, pp. 42082-42095. (**IF: 3.6**, at publication) — Apr. 2023

[J.6] **Hunmin Lee**, Mingon Kang, Donghyun Kim, Daehee Seo, Yingshu Li. Epidemic vulnerability index for effective vaccine distribution against pandemic. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 20, no. 6, pp. 3332-3342. (**IF: 3.93**, at publication) — Aug. 2022

[J.5] **Hunmin Lee**, Inseop Na, Kamoliddin Bultakov, Youngchul Kim. A BPR-CNN based hand motion classifier using electric field sensors. *Computers, Material and Continua*, vol. 71, no. 3, pp. 5413-5425. (**IF: 1.7**, at publication) — Jan. 2022

[J.4] Yoshitaka Inoue, **Hunmin Lee**, Tianfan Fu, Augustin Luna. drGT: Attention-Guided Gene Assessment of Drug Response Utilizing a Drug-Cell-Gene Heterogeneous Network. *BMC Bioinformatics* (**IF: 3.3**, at publication) — Accepted

[J.3] Jihun Bae, **Hunmin Lee**, Jinglu Hu. Sequential glioblastoma segmentation via Topological Data Analysis and spatial adjacency. *Biomedical Physics & Engineering Express* (**IF: 1.6**, at publication) — Feb. 2026

[J.2] Minjeong Seo, **Hunmin Lee**, Sehoon Kim Using Machine Learning in HRD Research: Applications to Advance AI-Driven Organizational Analytics. *Human Resource Development International* (**IF: 3.9**, at publication) — Jul. 2025

[J.1] Jihun Bae, **Hunmin Lee**, Jinglu Hu HeatGSNs: Enhancing Brain Tumor Segmentation and Classification with Eigenfilters and Learnable Low-Pass Graph Heat Kernels. *Biomedical Physics & Engineering Express* (**IF: 1.6**, at publication) — Dec. 2024

Hunmin Lee, Daehee Seo, Upper and Lower-Limb Motor Decoding for Adaptive and Generalized Neural Rehabilitation. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (Review round 2) — Under review

Hunmin Lee, Daehee Seo, Bridging Biosignals and Semantics: Cross-Modal Contrastive Learning between EMG and Text. *IEEE Journal of Biomedical and Health Informatics* (Review round 1) — Under review

Publications (Conferences)

[C.5] **Hunmin Lee***, Tanya George*, Eungjoo Lee. FedTTA: Enabling Robust Federated Test-Time Adaptation in Adversarial Weather for Distributed Surveillance, *The international society for optics and photonics (SPIE Defense and Security 2026)*, ***Oral*** (* Equal contribution) — Accepted

[C.4] **Hunmin Lee**, Ming Jiang, Qi Zhao. EMGCipher: Decoding Electromyography for Upper-limb Gesture Classification with Explainable AI for Resource Optimization, *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2024)*, ***Oral*** — Jul. 2024

[C.3] **Hunmin Lee**, Ming Jiang, Qi Zhao. FedAssist: Federated Learning in AI-Powered Prosthetics for Sustainable and Collaborative Learning, *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2024)*. — Jul. 2024

[C.2] **Hunmin Lee**, Mingon Kang, Yingshu Li, Daehee Seo, Donghyun Kim. Epidemic Vulnerability Index for Effective Vaccine Distribution Against Pandemic, *Bioinformatics Research and Applications: 17th International Symposium (ISBRA 2021)*, ***Oral*** — Nov. 2021

[C.1] **Hunmin Lee**, Sungil Jung, Young Chul Kim, Real-time hand motion frame extraction using electric potential sensors, *The 9th International Conference on Smart Media and Applications (SMA 2020)*. — Sep. 2020

Experience

Fasikl Incorporated, Research Engineer Intern — Bloomington, MN, USA; May. 2025 to current; 11 months

- Designed AI models for upper-limb neural rehabilitation & diagnosis of movement disorders (Parkinson's disease and essential tremor).

Research Assistant — 7.5 years

- University of Minnesota VIP Lab (Dr. Catherine Qi Zhao; Aug. 2022 - May 2025)
- Georgia State University (Aug. 2020 - Apr. 2022)
- Chonnam National University (Sep. 2018 - Apr. 2020)

Teaching Assistant — 10 academic semesters

- University of Minnesota: Machine Learning Fundamentals (CSCI 5521, 2023 Fall, 2024 Fall-2025 Fall), Introduction to Python (CSCI 1133, 2024 Spring), Introduction to Artificial Intelligence (CSCI 4511, 2023 Spring)
- Georgia State University: Theoretical Foundations of Computer Science (CSC 2510, 2022 Spring), Automata (CSC 4510, 2021 Spring)

Korea Power Electric Corporation, Software Engineer Intern — Naju, South Korea; Jun. 2020 to Aug. 2020

- Advanced Metering Infrastructure (AMI) software maintenance and deployment — 2 months

Projects

AI-powered Biomedical System for Rehabilitation (Supported by Fasikl Inc.) — Aug. 2024 to Current

- Designing an integrated machine learning model for Neural Computing Interface (NCI) for upper-limb rehabilitation
- Supported by Fasikl Incorporated

Efficient Learning for Distributed Network via Federated Learning (Supported by Google) — May 2025 to Current

- Designing a federated learning architecture that enables efficient and effective prototype-based learning for rapid adaptation in distributed network environments
- Supported by Google DeepMind (collaborator: Ms. Lucy Wang)

Distributed Learning for Object Detection Task under Adversarial Weather Conditions — Nov. 2025 to Current

- Designing a federated learning architecture that enables test-time adaptation in distributed environments under adversarial weather condition
- Collaboration with University of Arizona (Dr. Eungjoo Lee)

EAGER: Interpretable and Generalizable AI for Smart Manufacturing — Aug. 2022 to Jan. 2023, May 2023 to Aug. 2023

- Research and Development on an interpretable deep learning model for smart rehabilitation devices (prosthetics, wearable robotic devices)
- Supported by National Science Foundation (No. 2143197)

Global Research & Mentorship Program — Sep. 2021 to May. 2022

- Led a collaborative research project and mentored undergraduate students at Hanyang University in South Korea.
- Theme: Federated Learning Optimization in Non-IID Distributed Environments
- Supported by Hanyang University and Georgia State University.

Multi-user interface for IoT using non-contact capacitive sensing — Sep. 2018 to Apr. 2020

- Developed a mobile user interface architecture integrating non-contact capacitive sensing with an AI-based recognition system
- Supported by National Research Foundation, South Korea.

Teaching Experience

Machine Learning Fundamentals, CSCI 5521: Assigned for lab sessions implementing machine learning concepts using Python.
Organizer: University of Minnesota, Department of Computer Science and Engineering, 2023, Fall.

Theoretical Foundations of Computer Science, CSC 2510: Covered propositional and predicate logic, induction, recurrence relations, combinatorics, graph theory, and sets, relations, and functions, with applications in program verification, algorithm analysis, databases, and automata.
Organizer: Georgia State University, Department of Computer Science, 2022, Spring.

Awards and Honors

Doctoral Dissertation Fellowship (2026): Awarded $26,000 as a Doctoral Dissertation Fellow, recognizing the university's most accomplished Ph.D. candidates.
Organizer: University of Minnesota

IOP Outstanding Reviewer Award (2025): Recognized for high-quality peer review contributions to the Journal of Neural Engineering.
Organizer: IOP Publishing

2nd Prize, AI/ICT assistive device nationwide competition (Korea nationwide, 2024): Awarded 2nd prize for proposing ‘K-AI prosthetic model and platform architecture’.
Organizer: Korea Employment Agency for Persons with Disabilities, Korean Ministry of Employment and Labor

3rd Prize, Social Data Analysis nationwide competition (Korea nationwide, 2023): Awarded 3rd prize for proposing ‘GNN-based network analysis’ using Graph Convolution Network.
Organizer: Daishin Group, Korea ESG Lab

1st Prize, Industrial hemp breeding image and environmental data AI online nationwide hackathon (Korea nationwide, 2022): Awarded 1st prize for proposing ‘Korea hemp deep learning GNN model’ using GNN-based federated learning framework.
Organizer: Korean National Information Society Agency (NIA)

1st Prize, Georgia State University 3 Minute-Thesis award (2020): Awarded 1st prize for Georgia State University’s 3-minute thesis presentation.
Organizer: Georgia State University

1st Prize, Asia Open Data Challenge Contest (International, 2019): Developed Conv-LSTM model for wind-energy generation prediction model.
Organizer: GIGABYTE, Korea, Japan, Taiwan, Thailand

2nd Prize, Korea Electric Power Corporation Software Competition (Korea nationwide, 2019): Predicting solar energy generation amount using machine learning model.
Organizer: Korea Electric Power Corporation (KEPCO)

2nd Prize, University Student Idea National Contest (Korea nationwide, 2018): Awarded 2nd prize for nationwide university student contest for a given problem (Hyundai Motors vehicle mass-production challenge)
Organizer: Korean Ministry of Small and Medium-sized Enterprises and Startups, Hyundai Motors

Additional Activities

Academic Service (Journal Reviews) : IEEE Internet of Things Journal, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Mobile Computing, IEEE Transactions on Emerging Topics in Computing, IEEE Transactions on Computational Social Systems, IEEE Access, Scientific Reports (Springer), Journal of Neural Engineering (IOP Science), Engineering Research Express (IOP Science), Measurement Science and Technology (IOP Science) Computational Social Networks (Springer), Journal of Supercomputing (Springer), Journal of Clinical Medicine (MDPI), Sensors (MDPI), Mathematics (MDPI), Applied Sciences (MDPI), Electronics (MDPI), Biomedical Engineering / Biomedizinische Technik (De Gruyter, Germany)

Academic Service (Conference Reviews) : IEEE IPCCC 2021, IEEE ICCC 2021, MOTOR 2021, IEEE WCNC 2021, IEEE INFOCOM 2022

Military Service: Special Duty Team (SDT), Republic of Korea Army; Inje, South Korea, Mar. 2015 – Dec. 2016.

Volunteering: Volunteer lecturer teaching mathematics and English to visually impaired middle school students, Segwang School, Gwangju, South Korea, Mar. 2014 – Aug. 2014.

Technologies

Prgramming Languages: Python, C++, C, Java, SQL, MATLAB

Language: Korean (Native), English (Native-level)

Invited Talks

Sunchon National University: Research trends in AI-powered applications for neural decoding and rehabilitation (2025.05.13)

Sangmyoung University: Artificial Intelligence models for biomedical and rehabilitation engineering (2024.12.03)

Sunchon National University: Generative AI and their applications (2024.08.14)

Chonnam National University: Deep Neural Network and its applications in the biomedical field (2024.07.28)

Sangmyoung University: Graph neural networks and prototype learning (2024.06.15)

Sunchon National University: Designing interpretable deep learning model in biomedical engineering (2023.11.24)